

Python Assignment – 26

1: Write a Python program to implement a class named Demo with the following specifications:

- The class should contain two instance variables: no1 and no2.
- The class should contain one class variable named Value.
- Define a constructor (__init__) that accepts two parameters and initializes the instance variables.
- Implement two instance methods:
 - Fun() – displays the values of instance variables no1 and no2.
 - Gun() – displays the values of instance variables no1 and no2.

Create two objects of the Demo class as follows:

Obj1 = Demo(11, 21)

Obj2 = Demo(51, 101)

Call the instance methods in the given sequence:

Obj1.Fun()

Obj2.Fun()

Obj1.Gun()

Obj2.Gun()

Answer :

```
class Demo :
    Value = 0          #Class Variable

    def __init__(self,No1,No2):      #Constructor
        print("Inside Constructor")
        self.No1 = No1              #Instance Variable
        self.No2 = No2              #Instance Variable

    def Fun(self):                  #Instance Method
        print("Inside Fun() Instant Method")
        print("N01 : ",self.No1)
        print("N02 : ",self.No2)

    def Gun(self):                  #Instance Method
        print("Inside Gun() Instant Method")
        print("N01 : ",self.No1)
        print("N02 : ",self.No2)

def main():
    print("Class Demonstartion")

    #Create Object
    Obj1 = Demo(11, 21)
    Obj2 = Demo(51, 101)
```

```

# Instance Methods with call
Obj1.Fun()
Obj2.Fun()
Obj1.Gun()
Obj2.Gun()

if __name__ == "__main__":
    main()

```

2. Write a Python program to implement a class named Circle with the following requirements:

- The class should contain three instance variables: Radius, Area, and Circumference.
- The class should contain one class variable named PI, initialized to 3.14.
- Define a constructor (__init__) that initializes all instance variables to 0.0.
- Implement the following instance methods:
 - Accept() – accepts the radius of the circle from the user.
 - CalculateArea() – calculates the area of the circle and stores it in the Area variable.
 - CalculateCircumference() – calculates the circumference of the circle and stores it in the Circumference variable.
 - Display() – displays the values of Radius, Area, and Circumference.
- Create **multiple objects** of the Circle class and invoke all the instance methods for each object.

Answer :

```

class Circle:
    PI = 3.14

    def __init__(self):
        print("Inside Constructor")
        self.Radius = 0.0
        self.Area = 0.0
        self.Circumference = 0.0
        print()

    def Accept(self):
        print("Inside Radius")
        self.Radius = int(input("Enter the Radius of Circle : "))
        print("Your Entered Radius is : ",self.Radius)
        print()

    def CalculateArea(self):
        print("Inside Area")
        self.Area = self.PI * (self.Radius**2)
        print("Area of Circle is : ",self.Area)
        print()

```

```

def CalculateCircumference(self):
    print("Inside Circumference")
    self.Circumference = 2 * self.PI * self.Radius
    print("Circumference Of Circle is : ",self.Circumference)
    print()

def Display(self):
    print("Inside Display")
    print("Display.Radius : ",self.Radius)
    print("Display.Area : ",self.Area)
    print("Display.Circumference : ",self.Circumference)
    print()

def main():
    Obj1 = Circle()
    Obj1.Accept()
    Obj1.CalculateArea()
    Obj1.CalculateCircumference()
    Obj1.Display()

    Obj2 = Circle()
    Obj2.Accept()
    Obj2.CalculateArea()
    Obj2.CalculateCircumference()
    Obj2.Display()

    Obj3 = Circle()
    Obj3.Accept()
    Obj3.CalculateArea()
    Obj3.CalculateCircumference()
    Obj3.Display()

if __name__ == "__main__":
    main()

```

3: Write a Python program to implement a class named Arithmetic with the following characteristics:

- The class should contain two instance variables: Value1 and Value2.
- Define a constructor (__init__) that initializes all instance variables to 0.
- Implement the following instance methods:
 - Accept() – accepts values for Value1 and Value2 from the user.
 - Addition() – returns the addition of Value1 and Value2.
 - Subtraction() – returns the subtraction of Value1 and Value2.
 - Multiplication() – returns the multiplication of Value1 and Value2.

- Division() – returns the division of Value1 and Value2 (handle division by zero properly).
- Create multiple objects of the Arithmetic class and invoke all the instance methods.

Answer ;

```
class Arithmetic:
    def __init__(self, Value1, Value2):
        print("Inside Constructor")
        self.Value1 = Value1
        self.Value2 = Value2
        print()

    def Accept(self):
        print("Inside Accept")
        self.Value1 = int(input("Enter the Value1 Values :"))
        self.Value2 = int(input("Enter the Value2 Values :"))
        print("Value1 is : ", self.Value1)
        print("Value2 is : ", self.Value2)
        print()

    def Addition(self):
        print("Inside Addition")
        AddResult = self.Value1 + self.Value2
        print("Addition is : ", AddResult)
        print()

    def Substraction(self):
        print("Inside Substraction")
        SubResult = self.Value1 - self.Value2
        print("Substraction is : ", SubResult)
        print()

    def Multiplication(self):
        print("Inside Multiplication")
        MultResult = self.Value1 * self.Value2
        print("Multiplication is : ", MultResult)
        print()

    def Division(self):
        print("Inside Division")
        DivResult = self.Value1 / self.Value2
        print("Division is : ", DivResult)
        print()

def main():
```

```
Value1 = 0
Value2 = 0
Obj1 = Arithmetic(Value1,Value2)
Obj1.Accept()
Obj1.Addition()
Obj1.Substraction()
Obj1.Multiplication()
Obj1.Division()

Obj2 = Arithmetic(Value1,Value2)
Obj2.Accept()
Obj2.Addition()
Obj2.Substraction()
Obj2.Multiplication()
Obj2.Division()

if __name__ == "__main__":
    main()
```